Report Writing

Ozon Data Intelligence

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Abstract

This project implements a complete data science pipeline to analyze the electronics market on Ozon. We bypassed anti-bot protections to collect 16,000+ entries, performed rigorous cleaning on 10,000 rows, and conducted statistical audits. Using XGBoost, we achieved an R^2 of 0.89 in price prediction, identifying key market anomalies such as the "Apple Brand Tax" and category-based price inflation.

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1 Executive Summary

This report details an investigation into systemic pricing anomalies within the Ozon marketplace. The objective was to scrape, clean, and model hardware data to detect irregularities. Key findings include a massive price disparity between brands (Apple vs. Samsung) and the relative insignificance of RAM as a single pricing factor across mixed categories.

2 Data Collection Methodology

Scraping Ozon presented significant challenges due to aggressive anti-bot protections (Cloudflare).

- **Stealth Strategy:** We utilized `undetected-chromedriver` to mimic human browser behavior.
- **API Interception:** Instead of parsing raw HTML, we intercepted JSON responses from Ozon's internal `entrypoint-api`.
- **Dataset Scale:** 16,000+ raw rows were collected, featuring over 12 meaningful columns (Price, Brand, RAM, CPU Cores, Category, Rating, etc.).

3 Data Cleaning and Preprocessing

To ensure audit integrity, the raw data underwent a cleaning phase:

- **Deduplication:** Removed 6,000+ redundant entries caused by pagination overlap.
- **Missing Values:** Dropped records without price targets. Imputed missing specs using **Median** (for numericals like RAM) and **Mode** (for categoricals like CPU Brand).
- **Standardization:** Converted all price values to a unified numeric format.

4 Feature Engineering and Transformation

We transformed raw JSON noise into structured intelligence:

- **Flattening:** Nested technical specs were extracted into individual columns.
- **Encoding:** Categorical variables (Brand, Category) were processed via One-Hot Encoding.
- **Scaling:** Applied `StandardScaler` to continuous features to prepare for machine learning "interrogation."

5 Statistical Analysis & Hypothesis Testing

Using `scipy.stats`, we performed a formal audit of the market. Alpha level $\alpha = 0.05$.

6 Statistical Analysis & Hypothesis Testing (15%)

To uncover hidden patterns in the Ozon electronics market, our team conducted a multi-layered statistical audit using `scipy.stats`. The significance level was set at $\alpha = 0.05$.

6.1 H1: Brand Premium (Apple vs. Samsung)

Null Hypothesis (H_0): There is no significant price difference between Apple and Samsung.
Test Results: $t = 15.0675, \mathbf{p < 0.001}$.
Verdict: REJECTED H_0. Apple prices are statistically significantly higher than Samsung, confirming a brand-based price anomaly.

6.2 H2: Impact of RAM on Price

Null Hypothesis (H_0): RAM capacity has no significant impact on device price.
Test Results: Correlation $r = -0.0711, \mathbf{p = 6.72 \times 10^{-13}}$.
Verdict: REJECTED H_0. While significant, the negative correlation indicates that RAM is not a straightforward price driver across mixed categories.

6.3 H3: Price Equality Across Categories (ANOVA)

Null Hypothesis (H_0): Mean prices across Laptops, Smartphones, and Tablets are equal.
Test Results: $F = 3013.7925, \mathbf{p < 0.001}$.
Verdict: REJECTED H_0. At least one category has a statistically significant difference in mean price.

6.4 H4: Comparative Audit: Laptops vs. Smartphones

Null Hypothesis (H_0): Mean price of Laptops is equal to the mean price of Smartphones.
Test Results: $t = 50.4362, \mathbf{p < 0.001}$.
Data Audit:

- Mean Laptop Price: 536,403.23
- Mean Smartphone Price: 111,561.94
- Difference: 424,841.29

Verdict: REJECTED H_0. There is a statistically significant difference in mean prices between laptops and smartphones.

7 Predictive Modeling

We built an "Auditor Model" to predict fair market value based on specifications.

- **Algorithm:** XGBoost (Gradient Boosting).
- **Performance:** Achieved $R^2 = 0.89$ and low RMSE.

- **Baseline:** Compared against Random Forest ($R^2 = 0.82$), proving that XGBoost better captures non-linear brand premiums.

Table 1: Model Performance Comparison

Model	MAE	RMSE	R² Score
Random Forest	Moderate	Moderate	0.82
XGBoost	**Very Low**	**Low**	**0.89**

8 Final Insights and Recommendations

- **Recommendation 1:** Implement automated ML auditing to flag products where price deviates $> 20\%$ from the predicted hardware value.
- **Recommendation 2:** Consumers should prioritize category-based research, as "Brand" is a more expensive feature than "Performance."

9 Conclusion

The pipeline successfully bypassed marketplace protections to generate a robust 10,000-row dataset. Our analysis proves that while branding creates significant price anomalies, machine learning can effectively predict market value with high precision.